Memory Is Not Enough: Separating Drift from Background Conflict in Continual Document Understanding

Supplementary Material

Roadmap. This appendix holds the detail behind every claim of the main paper. Appendix A gives the datasets, the exact session composition of each scenario and the clean protocol in full. Appendix B defines the three ways we score a class-incremental session, works one receipt through them, and compares session-restricted AA with the mask-free full-label F1. Appendix C describes CoLaR, SBR and SBR-Recall at the level of the code, with the supervision plan of every (session, block) pair, pseudo-code and memory accounting. Appendix D gives the retention matrices, the per-seed values behind every headline cell, the LexSlot stacking comparison, SBR-Recall on the mixed stream and the delta-capacity ablation. Appendix E collects the evidence that the background conflict sits in the labels and not in the memory; Appendix F does the same for drift and holds the falsification ledger, the memory-content ladder and the image comparison. Appendix G documents the baselines, our ports and the settings of every legacy-protocol (†) cell, and Appendix H lists what is needed to reproduce the runs. Every number in the generated tables (file names `supp_*.tex`, `tab_*.tex`) is read from the run directories by `scripts/rca/make_clean_tables.py`; a † marks a legacy-protocol source, and "gate", "rescore" and "grid" name the three groups of legacy runs described in Appendix G.

A. Setup details

Datasets. FUNSD (`nielsr/funsd-layoutlmv3`; 149/50 train/test forms), CORD (`naver-clova-ix/cord-v2`; 800/100/100 receipts), SROIE (`mp-02/sroie`; 626/347 receipts). We use the dataset-provided OCR and boxes throughout, page images at 224×224 and a text length of 512 tokens; with the 197 image tokens of LayoutLMv3 a document is a sequence of 709 positions.

CIL-CORD. CORD's 30 fine-grained classes (61 BIO tags with "O") arrive in five sessions of six classes (Tab. 1). A session's training and test pools keep every receipt that contains at least one of that session's entities, and every token of a class outside the session is labelled "O". The head grows by twelve tags per session. Joint training uses the full-label pool (each receipt once, with all its labels) rather than the concatenation of the masked per-session copies, which would feed contradictory labels for the same document. Class imbalance across sessions is left as it is.

Table 1. **CIL-CORD sessions** (`_CORD_CANONICAL_SESSIONS` in `doccl/data/scenarios.py`). Pool: receipts of the CORD training split that contain at least one of the session's entities; under the clean protocol 10% of each pool is held out for early stopping (the per-epoch step counts at batch size 2 in the clean logs, 360/172/247/326/351, match 90% of the pools).

Session	Classes (prefix `menu.`, `sub_total.`, `total.` as listed)	Pool
1	menu: cnt, discountprice, itemsubtotal, nm, num, price	800
2	menu: unitprice, vatyn, sub.cnt, sub.nm, sub.price, sub.unitprice	382
3	sub_total: discount_price, etc, othersvc_price, service_price, subtotal_price, tax_price	549
4	total: cashprice, changeprice, creditcardprice, emoneyprice, menuqty_cnt, menutype_cnt	723
5	total.total_etc, total.total_price, void_menu.nm, void_menu.price, sub.nm, sub.cnt	780

Table 2. **Mixed sessions** (`build_mixed`). Schema: the dataset whose annotation scheme the session uses, which SBR's supervision rule reads (Appendix C.2). Pool: training documents, read from the per-epoch step counts at batch size 2 of `mixed_colar_sbr2_seed42` (no held-out split), so ±1 where the last batch holds one document.

Session	Dataset	Tags introduced	Schema	Pool
1	FUNSD	HEADER, QUESTION	funsd	149
2	FUNSD	ANSWER	funsd	147–148
3	SROIE	COMPANY, DATE, ADDRESS, TOTAL	sroie	625–626
4	CORD (super)	menu, sub_total	cord	799–800
5	CORD (super)	total, void_menu, sub	cord	797–798
6	FUNSD	none (all FUNSD labels)	funsd	149

Mixed. Six sessions over three datasets and two annotation schemas per form type (Tab. 2): two class-incremental FUNSD sessions, a domain shift to SROIE, two class-incremental sessions on CORD's super-classes, and a return to FUNSD with all its labels. The head grows to 25 tags; the FUNSD revisit adds none. Boundaries therefore alternate between new classes within a dataset and a new dataset, and the last session checks retention of the first.

DIL unified schema. A single four-entity label space {HEADER, KEY, VALUE, OTHER} (9 BIO tags) is held fixed over FUNSD→SROIE→CORD. FUNSD: HEADER→HEADER, QUESTION→KEY, ANSWER→VALUE. SROIE: COMPANY→KEY; ADDRESS/DATE/TOTAL→VALUE. CORD: menu/sub_total/total→VALUE; void_menu/sub→OTHER. KEY originates only from FUNSD and SROIE, so the schema is VALUE-dominant; we report per-class F1 for every DIL result and treat a high aggregate as retention only if the sparse classes are retained. The reverse order is used

as a robustness check.

Backbones. LayoutLMv3-base is the primary encoder (text, layout and image patches in one sequence). LiLT and BROS are text-and-layout encoders and BERT-base is text only; all four have twelve layers of width 768. Every head is a single linear layer: an MLP head with ten or more labels saturates and collapses to all-background after class-incremental expansion. When the head grows, old rows are kept and new rows are drawn from $\mathcal{N}(0, 0.02^2)$.

The clean protocol in full. Every clean run uses fp32 (`training.amp=false`), batch size 2 with gradient checkpointing, AdamW with learning rate 5×10^{-5}, weight decay 0.01 and gradient clipping 1.0. Each session trains for at most 10 epochs (100 for joint training) and stops early when the F1 on a held-out split has not improved by 0.1 points for 2 epochs, after which the best weights are restored. The held-out split is 10% of that session's *training* pool, drawn with a generator seeded by the run seed (`split_val` in `scripts/train.py`); the test split is never used for model selection. Class-incremental sessions are scored with session-restricted scoring (Appendix B); on CIL-CORD the final model is also scored on the unmasked test set (full-label F1). Mixed and DIL have no full-label set, so the column is CIL-CORD only. Seeds are 42, 7 and 123. CoLaR-family runs use k=4, rank r=128, 50 stored documents per session and 4 replayed documents per step; the clean ER row keeps 250 raw documents, as many as CoLaR stores on CIL-CORD. A table cell switches to the clean protocol only when all three clean seeds exist, so no cell mixes the protocols.

Compute. Every run fits a single GPU; the full generality grid is roughly 200 GPU-hours on a 24 GB card. Per-run times of the headline cells are in Tab. 6 and Appendix H.

B. Scoring, explained

Definitions. Let $\mathcal{A}_j$ be the tags of session j and $\mathcal{D}_j$ its test documents, whose gold labels show only $\mathcal{A}_j$ (every other entity is "O"). All three scores are entity-level F1 from `seqeval`.

- *Legacy scoring* compares the model's raw predictions on $\mathcal{D}_j$ with the masked gold labels.
- *Session-restricted scoring* first maps every predicted tag outside $\mathcal{A}_j$ to "O" and then compares. It does not re-take the argmax among the tags of $\mathcal{A}_j$: a token whose best tag belongs to another session becomes "O", never its second-best in-session tag. For SBR the mapping is applied after the block decision of Eq. (1).
- *Full-label F1* scores the final model, with no mapping, on the unmasked CORD test set (all 100 receipts, all 30 classes).

Table 3. **Session-restricted AA and full-label F1 of the same final model**, every clean-protocol run on disk that records both (CIL-CORD, LayoutLMv3; generated, `supp_agree.tex`).

Scenario	Method	Seed	AA	Full-label F1	Difference
CIL-CORD	CoLaR-SBR	42	87.8	88.8	+1.1
CIL-CORD	CoLaR-SBR	7	91.3	91.2	−0.1

A worked example. Take one receipt with a menu item name (class `menu.nm`, session 1) and the total price (`total.total_price`, session 5), and a model that labels both correctly. In session 1's test labels the total price is "O", so under legacy scoring the correct total-price span is a false positive there: precision 1/2, recall 1, F1 0.67 for session 1 on this receipt. Symmetrically, the correct menu-name span is a false positive in session 5. A perfect model is charged once in every session that does not own a predicted entity, so legacy scoring penalises the model for knowing more classes, and the penalty grows with the number of sessions; this is why joint training, which labels every class, reads 33.3 under legacy scoring on CIL-CORD (Tab. 18) and 91.9±1.3 under restricted scoring. Under session-restricted scoring the total-price prediction is mapped to "O" in session 1 and counts in session 5 only, so both sessions score 1. If instead the model labels the total price as `menu.price` (a session-1 class), the predicted span is a false positive in session 1 and the gold total is a miss in session 5: one false positive and one false negative, exactly what ordinary entity F1 charges for a wrong type.

Why restricted scoring charges each error at most once. Every predicted entity has one class, and that class belongs to exactly one session; outside that session the mapping removes it, so it can be a false positive only in its owning session. Every gold entity appears unmasked only in the test labels of its own session, so it can be a false negative only there. A confusion between sessions therefore still costs a miss. The caveat is coverage: $\mathcal{D}_j$ keeps only receipts that contain an entity of session j, so a false positive of a session-j class on a receipt without any session-j entity is never counted. Full-label F1 has no such gap, which is why we report it.

The two metrics agree. Tab. 3 lists every clean-protocol run that records both numbers. For the CoLaR-SBR runs listed the two differ by about one point or less on each seed, so the restricted AA of the main comparison (Tab. 3 of the main paper) is not inflated by the coverage gap. The table is regenerated as clean runs complete.

C. Method details

C.1. CoLaR

What is stored. After the first session, the embeddings and layers $1..k$ are frozen (`_apply_freeze_map` in `latent_replay.py`); everything above, and the head, stays plastic. At the end of each session the hidden state entering layer $k{+}1$ is captured for 50 training documents of that session (the first 50 of the shuffled loader), from the same frozen layers that will produce it forever after, so a stored activation cannot go stale. Each document's activation matrix $H_j \in \mathbb{R}^{709\times768}$ is factorised by a truncated SVD and kept as $(U_r\Sigma_r) \in \mathbb{R}^{709\times r}$ and $V_r^\top \in \mathbb{R}^{r\times768}$ in fp16, together with the document's boxes, attention mask and labels (and, for SBR, the index of the session that stored it). No raw pixels and no token ids are kept.

Replay. At every training step four stored documents are sampled uniformly, $\hat{H}_j = (U_r\Sigma_r)V_r^\top$ is rebuilt, and a forward pre-hook on layer $k{+}1$ replaces the hidden state of a dummy input by $\hat{H}_j$; the stored boxes and mask drive the layout position biases and the attention mask of the replayed pass. The loss is the cross-entropy of the current batch plus the cross-entropy of the replayed documents with their stored labels, with weight 1. Gradient checkpointing is switched off for the replayed pass, because recomputation would re-run layer $k{+}1$ on the dummy input.

Memory accounting. `memory_bytes` counts, per stored document, the two fp16 factors, $2r(709{+}768)$ bytes (378,112 at $r{=}128$), and the boxes, mask and labels at the eight bytes per entry in which they are held (24,576 bytes); SBR adds one byte for the session index. A document therefore costs 402,689 bytes in CoLaR-SBR, and the 250 documents of CIL-CORD give the 100,672,250 bytes recorded as `replay_memory_bytes` in every CoLaR-SBR run (101 MB; 1 MB $= 10^6$ bytes), against 272.3 MB for the same activations uncompressed (training log). On CIL-CORD and Mixed the stored record also carries the document's labels in the final label space, which only the true-label refresh diagnostic of Appendix E reads; CoLaR and SBR never read it, and the count above does not include it (4,096 bytes per document).

C.2. SBR

Blocks. A block is created by every session that introduces at least one tag not yet covered by a block; it holds those new tags $\mathcal{T}_b$, a background weight vector w_b (a linear layer from the 768-dimensional head input to one logit) and the schema of its session. A session that only re-annotates known tags (every DIL session after the first, the Mixed FUNSD revisit) creates no block, so on DIL SBR has exactly one block and differs from CoLaR only in its separate background logit and the thresholded decision. The tag logits z are the rows of the one shared linear head.

Table 4. **Supervision plan** of a document of session s (rows) for block b (columns): F = full, N = all-background, – = skipped. Derived from `supervision_plan`; the plan of each current session is also printed in the training logs (`clean_2026-11/cil_cord_colar_sbr_seed42`, `gates/.../mixed_colar_sbr2_seed42`) and matches. Entries for blocks created after session s apply when s's stored documents are replayed.

CIL-CORD						Mixed					
$s\backslash b$	1	2	3	4	5	$s\backslash b$	1	2	3	4	5
1	F	–	–	–	–	1 FUNSD	F	–	N	N	N
2	–	F	–	–	–	2 FUNSD	–	F	N	N	N
3	–	–	F	–	–	3 SROIE	N	N	F	N	N
4	–	–	–	F	–	4 CORD	N	N	N	F	–
5	–	–	–	–	F	5 CORD	N	N	N	–	F
						6 FUNSD	F	F	N	N	N

The supervision rule. For a document of session s with tag set $\mathcal{A}_s$ and schema d_s, `supervision_plan` assigns every existing block b one of three modes:

- **full** if $\mathcal{T}_b \subseteq \mathcal{A}_s$: session s annotates every tag of the block, so its "O" labels are true negatives for that block; the target is y for tags of $\mathcal{T}_b$ and O_b for every other token, including annotated entities of other blocks;
- **all-background** if $d_b \neq d_s$: the block belongs to another annotation schema, none of its entities can occur in the document, and every token's target is O_b;
- **skipped** otherwise: same schema, block not annotated, so the document's "O" says nothing about it.

A stored document is supervised by the plan of the session that stored it, evaluated over the blocks that exist when it is replayed. Tab. 4 spells the plans out. On CIL-CORD every session has the same schema, so a document trains only its own block and the all-background mode never occurs. On Mixed the all-background mode is what keeps a block from claiming tokens of another dataset. An earlier version that created a block for every session and had no all-background mode failed on DIL exactly there: the receipt session reached 23.6 F1 on its own diagonal, because the FUNSD block, never trained on receipts, claimed receipt tokens (`dil_colar_sbr_seed42`, AA 66.5).

Inference. For each token every block reports $e_b = 1 - p_b(\mathrm{O}_b \mid x)$; the block with the largest e_b labels the token with its most probable tag if $e_b > \tau$ ($\tau{=}0.5$), and the token

Algorithm 1 CoLaR-SBR: one session s (tags $\mathcal{A}_s$, schema d_s)

$\mathcal{N} \leftarrow \mathcal{A}_s \setminus \bigcup_b \mathcal{T}_b$
if $\mathcal{N} \neq \emptyset$ **then** add block ($\mathcal{T} = \mathcal{N}$, w, d_s)
end if
hold out 10% of the session's training pool
for epoch $= 1..10$, until early stopping **do**
  for each minibatch X of session s **do**
    $\mathcal{L} \leftarrow \sum_{(b,\text{mode})\in\text{plan}(s)} \text{CE}_b(X, \text{mode})$
    sample 4 stored documents; rebuild $\hat{H} = (U_r\Sigma_r)V_r^\top$
    inject $\hat{H}$ at layer $k{+}1$; for each, add $\sum_{\text{plan}(s')} \text{CE}_b$ of its session s' (mean over the four) to $\mathcal{L}$
    AdamW step on layers $k{+}1..12$, head, all w_b
  end for
end for
restore the best weights; **if** $s = 1$: freeze embeddings and layers $1..k$
store 50 documents: $(U_r\Sigma_r, V_r^\top)$, boxes, mask, labels, s

is "O" otherwise (ties go to the lower block index):

$$\hat{y}(x) = \begin{cases} \arg\max_{t\in\mathcal{T}_{b^\star}} p_{b^\star}(t \mid x) & e_{b^\star} > \tau, \\ \text{O} & \text{otherwise}, \end{cases} \qquad (1)$$
$$b^\star = \arg\max_b e_b.$$

No task identity is used, and the session restriction of the evaluation is applied only after this decision.

C.3. SBR-Recall

Deltas. Every linear layer named query, key, value or dense in layers $k{+}1..12$ is wrapped by a module that holds one low-rank pair (A_b, B_b) per delta and adds $\frac{\alpha}{r}\, xA_b^\top B_b^\top$ to the base output when delta b is active. "Dense" covers the attention output projection and both feed-forward layers, so a layer carries six deltas and the eight layers above $k{=}4$ carry 48. A is initialised Kaiming-uniform and B at zero, so a fresh delta is the identity. The headline setting is rank 64, $\alpha{=}128$; the capacity ablation uses rank 8, $\alpha{=}16$ on query, key and value only.

Which delta a session trains. The first session trains the base network itself; afterwards the whole encoder is frozen (θ_0) and only the head, the background weights and the current delta are trainable. A session that creates a block gets a fresh delta; a session that creates none reuses the delta of the first block it trains fully (on the Mixed revisit this is the base of the first session, so that session trains only the readout).

Losses under the delta a block is read under. `_loss_under_deltas` groups the blocks of a document's plan by the delta each block is read under, runs one forward pass per distinct delta, and computes each block's term from the pass under its own delta. Frozen deltas add only the head to the autograd graph. Stored documents, when there is a memory, are grouped by session and pass through the layer-k injection under each delta in the same way. An earlier implementation computed the all-background terms of other-schema blocks under the *current* delta, which trained those blocks' readout on features from a trunk state they are never read under; its Mixed run (`mixed_colar_sbr_recall_r64_seed42`) is an implementation error, not a result, and is not reported. CIL-CORD has a single schema, so its runs never contain such terms and are unaffected.

Algorithm 2 SBR-Recall: what changes relative to Algorithm 1

if $s = 1$ **then** train the base as in Algorithm 1; then freeze the encoder (θ_0)
else if session s adds a block **then** create Δ_s ($B{=}0$); trainable: Δ_s, head, w_b
else reuse the delta of the first block s trains fully
end if
loss: for each distinct delta Δ among the plan's blocks, forward X under $\theta_0{+}\Delta$ and add the CE of every block read under Δ
inference: forward under each distinct delta; block b uses the pass under its own delta; decide by Eq. (1)

Inference and cost. One forward pass per distinct delta produces the logits of the blocks read under it, and Eq. (1) combines them. Our implementation repeats the full forward pass for each delta: five passes on CIL-CORD (base plus four deltas). Since the embeddings and layers $1..k$ are identical in all of them, a cached implementation repeats only the $12 - k$ upper layers, $(k + 5(12 - k))/12 \approx 3.7$ encoder-layer equivalents of one forward pass at $k{=}4$, which is the figure in the main paper.

Parameters and memory. A rank-64 delta has $8 \times [3 \cdot 64 \cdot (768{+}768) + 64 \cdot (768{+}768) + 2 \cdot 64 \cdot (768{+}3072)] = 7{,}077{,}888$ parameters; `memory_bytes` counts the deltas at two bytes per parameter (they are trained in fp32), so the four deltas of CIL-CORD give 56,623,104 bytes (57 MB), as recorded in every rank-64 run. A rank-8 delta on query, key and value has 294,912 parameters (2.4 MB for four). With the 50-document memory on Mixed the run records 177 MB, the deltas plus the CoLaR store.

D. Extended results

D.1. Retention matrices

Tab. 5 gives $R[i, j]$, the F1 on session j after training session i, averaged over the seeds named in each title; the sources

follow the same rule as the main tables, so a matrix moves to the clean protocol with its table cell. The matrices are the numbers behind the retention figure of the main paper. With a shared softmax every column decays as soon as the next session arrives, and later sessions are learned poorly even on their own diagonal, which is the current-side half of the conflict. With the blocked readout every session is learned and its column loses a few points over the following sessions. Under SBR-Recall every column stays flat. On Mixed, in the runs shown, CoLaR-SBR keeps every session close to its diagonal and the FUNSD sessions rise again at the revisit.

D.2. Per-seed values behind the headline cells

Tab. 6 lists, for every cell of the main comparison and of the 2×2, the seeds of the source the cell is read from and the mean training time per run. Every "within", "at" and "on par" statement in the main paper is checked against these per-seed values, not against the mean.

D.3. Stacking LexSlot on CoLaR-SBR

LexSlot adds extra head slots that a session updates only when the OCR vocabulary of its documents selects them; it stores the token ids of its documents, unlike CoLaR. LexSlot exists only as fp32 gate runs, so Tab. 7 compares it with the CoLaR-SBR gate runs of the same protocol, never with the main-table source. Stacked on CoLaR-SBR it moves the per-seed AA by a few points in either direction and leaves the mean within two points on every scenario, and it narrows the spread between seeds on CIL-CORD (88.0 ± 2.0 against 89.7 ± 0.5) and DIL (86.2 ± 1.3 against 87.0 ± 0.4). It stays below SBR-Recall on CIL-CORD. We read it as a second partition of the same readout that SBR already partitions, and keep it as a row, not a method.

D.4. SBR-Recall on the mixed stream

Three Mixed runs of SBR-Recall exist, all rank 64 and seed 42. The first (`_r64`) computed the cross-schema terms under the wrong delta (Appendix C.3) and is not a result. The second (`_r64b`) is the corrected method without a memory; the third (`_r64c`) adds the 50-document CoLaR memory. Their matrices are in Tab. 8.

Without a memory the blocks see-saw. Read the no-memory matrix by columns. FUNSD sessions 1 and 2 are learned (87.6, 90.8) and fall to 0.0 as soon as SROIE arrives in session 3; SROIE is learned (82.0) and falls below 10 when CORD arrives; both CORD sessions hold (95.8, 95.5) until the FUNSD revisit, when FUNSD returns (87.3, 85.6) and SROIE and the CORD blocks fall to 8.1 or below. The mechanism is the all-background mode. On a current document of another schema every old block is trained towards O_b on every token, and without stored documents nothing trains it towards its own tags, so its background logit wins and the block stops claiming tokens. The trunk is frozen and the old deltas are not updated, so this is a readout effect, not drift: the readout rows and background weights of the old blocks are shared across deltas and are moved by these terms. When a block's own schema returns, it is supervised fully again and recovers within one session.

With the memory the see-saw disappears. Stored documents supply the positive counterweight: in the with-memory matrix no column falls to zero, the CORD blocks hold above 96, and the run reaches 85.9. The weakest cell is FUNSD session 2 (46–48 through the SROIE and CORD sessions, 75.0 after the revisit). Across schemas, isolation by deltas does not replace the memory, and CoLaR-SBR (89.7 in the main table) remains the better method on Mixed; one seed.

D.5. Delta capacity

Tab. 9 gives the diagonal and the final row of every SBR-Recall capacity run on CIL-CORD. Without a memory the final row is within 1.5 points of the diagonal at either rank, so isolation retains what was learned. At rank 8 on query, key and value, however, sessions 3 and 4 (the sub-total and total fields, mostly prices) are learned only to between 61 and 68 F1 on the diagonal, and adding the memory does not help (AA 80.5 against 82.5). At rank 64 with the attention output and feed-forward layers added, those two diagonals reach 88.8 to 96.2 on all three seeds. Isolation was sound at rank 8; the capacity of the delta was the limit.

E. Analysis of the background conflict

Takeaway. Repairs that change only the labels the memory and the current documents supervise move CIL-CORD AA from about 30 to about 60 without changing what the memory stores; a session-blocked readout with the same memory moves it to 88.0; and the readout without a memory, or on a frozen trunk, does not. The conflict is in the labels, and its remedy is in the readout. Tab. 10 lists the runs with their seeds.

Stale labels in the memory. A document stored at session t labels every entity of a later session "O". Refreshing the memory with the true labels at each boundary rewrites many stored tokens: on Mixed, 3120–3307, 2640–2932, 0 and 305–316 tokens at the four refreshes (50, 100, 150 and 200 stored documents; three seeds, earlier grid, legacy scoring). A deployable variant that refreshes with the model's own confident predictions rewrites none on seeds 42 and 7 and $7/0/0/36$ tokens on seed 123: the model is confident on almost no stale-"O" token, so the pseudo-label refresh is a no-op (falsification ledger, Tab. 12). Under restricted scoring the true-label refresh reaches 61.2 on CIL-CORD (one seed);

Table 5. **Retention matrices** $R[i, j]$ (row i: after training session i; column j: session j; session-restricted scoring, LayoutLMv3), mean over the seeds in each title; †: legacy protocol (Tab. 17). Generated (`supp_mat_*.tex`).

CoLaR, shared softmax, CIL-CORD (seeds 42/7, rescore†)

$i\backslash j$	1	2	3	4	5
1	94.7				
2	48.1	82.0			
3	49.6	54.4	37.1		
4	13.1	28.3	66.3	34.0	
5	14.1	24.4	53.8	40.9	18.6

CoLaR-SBR, CIL-CORD (seeds 42/7/123, gate†)

$i\backslash j$	1	2	3	4	5
1	94.2				
2	92.3	94.1			
3	91.4	86.2	94.1		
4	90.9	84.3	89.8	89.4	
5	90.7	79.3	87.7	86.4	96.0

SBR-Recall, CIL-CORD (seeds 42/7/123, gate†)

$i\backslash j$	1	2	3	4	5
1	94.2				
2	94.4	91.6			
3	94.4	92.0	93.7		
4	94.4	92.1	93.8	90.4	
5	94.4	92.1	94.3	90.3	93.0

CoLaR-SBR, Mixed (seeds 42/7, rescore†)

$i\backslash j$	1	2	3	4	5	6
1	86.2					
2	87.0	90.7				
3	86.5	88.7	82.3			
4	85.0	86.2	79.6	96.3		
5	85.4	90.0	78.6	95.4	96.0	
6	89.6	91.6	77.9	95.1	93.6	90.4

it is not deployable, since it reads labels that a continual learner never sees.

Stale labels in the current documents. The true-label refresh still leaves the newest session weak on its own diagonal (31.8, 14.4 and 23.0 for sessions 3 to 5 in `cil_cord_colar_r_oracle_seed42`), because current documents label every older entity "O". Slicing only the replay loss to the classes alive at storage gives 42.0 on average over two seeds; slicing the current loss as well (an "O" token of the current batch is trained only against O and the new classes) reaches about 60 on average over three seeds, with a 20-point spread between seeds. Adding pseudo-labels on top, in one direction or both, does not help (48.0 and 44.1, one seed each). Each of these acts on the shared background logit, and each leaves it pulled in two directions on look-alike price fields.

The readout without the memory, and the memory without plasticity. SBR without a memory scores 20.4 (one seed): the labels are consistent within each block, but the trunk drifts and old blocks fall to 0 to 5 F1 by the last session (Tab. 10; `cil_cord_colar_sbr_noreplay_seed42`). With every encoder layer but the last frozen after the first session and no memory, SBR scores 14.8 (one seed): sessions 3 to 5 are learned poorly or not at all (4.3, 22.0 and 0.0 on the diagonal) while session 2 is learned at 87.0 on the same frozen trunk. A likely reading is that old blocks out-vote the new ones on surface-identical prices unless the trunk can adapt; the run does not separate this from a block that cannot fit on a frozen trunk. Both plasticity and the memory are needed, and the blocked readout is what lets them combine.

Readout calibration. What SBR still loses to joint training is not a calibration error of the block competition. On the fp16 rescore run of CoLaR-SBR (CIL-CORD, seed 42, AA 85.2), the best global threshold gives 85.4 and a threshold plus a per-block bias tuned on the test set itself gives 85.8, so calibration is worth at most 0.6 (one seed). On DIL, with one block, tuning the threshold gains at most 0.7 and the test-set oracle at most 1.1. The remaining loss is retention of early sessions (Tab. 5), which SBR-Recall removes.

F. Analysis of drift

F.1. Displacement by depth

Takeaway. The classifier is where old-task-weighted parameters move most, but that ratio is inflated by construction; inside the encoder, weighted displacement sits in the early and middle layers, so "late layers" is a statement about rep-

Table 6. **Per-seed AA of every headline cell** (LayoutLMv3; session-restricted scoring on CIL-CORD and Mixed). Source: clean = `results/clean_2026-11`; gate = `results/gates/cil_protocol_2026-11`; rescore = `results/rescore`; grid = the earlier `results/<run>` grid; †: legacy protocol (settings per row in Tab. 17). [c]: an fp16 run that collapsed to all-background, excluded from the LwF mean. h/run: mean training-loop time per run from `metrics.json`; precision and hardware differ between groups, so the column is a record, not a cost comparison. Generated (`supp_perseed.tex`).

Row	Source	42	7	123	h/run
CIL-CORD					
Naive	rescore†	19.0	19.1	18.6	1.3
SBR, no memory	gate†	20.4	–	–	1.5
CoLaR	rescore†	28.8	31.9	–	1.4
CoLaR-SBR	gate†	87.6	86.3	90.1	2.4
SBR-Recall	gate†	92.7	93.4	92.4	1.5
Joint training	rescore†	92.8	90.5	92.4	0.4
ER	rescore†	26.9	22.0	24.0	3.4
CoLaR-SBR + LexSlot	gate†	90.2	89.7	89.2	2.3
LwF	rescore†	18.4	18.5	0.0[c]	1.3
DER++	rescore†	21.3	18.8	21.1	8.2
CODA-Prompt	rescore†	14.0	15.9	–	4.0
CL-LoRA (2025)	rescore†	18.7	18.6	–	3.0
CPFD (2023)	rescore†	2.7	0.0	–	1.0
IS3 (2024)	rescore†	5.6	7.7	–	2.2
LCA (2026)	rescore†	13.7	7.0	–	1.6
Mixed					
Naive	rescore†	44.4	43.7	–	0.7
CoLaR	rescore†	75.6	74.6	–	1.5
CoLaR-SBR	rescore†	89.8	89.6	–	1.4
Joint training	rescore†	86.9	88.5	–	0.8
ER	rescore†	75.6	73.3	–	2.8
CoLaR-SBR + LexSlot	gate†	88.5	90.0	90.3	2.5
SBR-Recall	gate†	85.9	–	–	5.9
LwF	rescore†	47.7	0.0[c]	–	1.1
DER++	rescore†	75.1	74.0	–	4.3
CODA-Prompt	rescore†	12.1	10.6	–	4.4
CL-LoRA (2025)	rescore†	45.7	43.9	–	2.2
CPFD (2023)	rescore†	0.3	15.1	–	0.8
IS3 (2024)	rescore†	0.0	40.6	–	1.6
LCA (2026)	rescore†	38.0	36.4	–	1.3
DIL					
Naive	grid†	40.1	43.0	40.9	0.5
CoLaR	grid†	87.6	88.3	87.4	1.5
CoLaR-SBR	gate†	86.8	84.8	87.1	1.5
Joint training	grid†	89.7	89.0	87.5	0.3
ER	grid†	86.2	88.7	88.6	0.2
CoLaR-SBR + LexSlot	gate†	87.2	87.2	86.6	1.4
LwF	grid†	42.5	44.6	43.8	0.3
DER++	grid†	88.2	87.1	88.8	0.2
CODA-Prompt	grid†	33.4	31.1	33.0	2.2
CL-LoRA (2025)	grid†	40.8	41.3	41.3	1.6
CPFD (2023)	grid†	0.0	4.2	14.2	0.8
IS3 (2024)	grid†	38.3	–	–	1.2
LCA (2026)	grid†	46.4	43.8	41.5	1.0

resentations only.

For a checkpoint pair $(\theta^{(t-1)}, \theta^{(t)})$ and a parameter group g we compute the old-task-Fisher-weighted displacement $D_g = \sum_{\phi \in g} F_\phi^{(t-1)} (\theta_\phi^{(t)} - \theta_\phi^{(t-1)})^2$. Tab. 11 reports it at the first task boundary, mean over the live seeds of each condition, and Fig. 1 shows the profile. The classifier's

Table 7. **CoLaR-SBR / + LexSlot (difference)**, AA per seed; both sides are fp32 gate runs† (one protocol per row; Mixed CoLaR-SBR has one gate seed). Generated.

Scenario	Protocol	42	7	123
CIL-CORD	gate†	87.6 / 90.2 (+2.5)	86.3 / 89.7 (+3.5)	90.1 / 89.2 (−0.9)
Mixed	gate†	89.7 / 88.5 (−1.2)	– / 90.0	– / 90.3
DIL	gate†	86.8 / 87.2 (+0.5)	84.8 / 87.2 (+2.5)	87.1 / 86.6 (−0.5)

Table 8. **SBR-Recall on Mixed** (rank 64, seed 42, fp32 gate runs; session-restricted scoring), without and with the 50-document memory. Sessions as in Tab. 2. Generated (`supp_mat_recall_mix_*.tex`).

SBR-Recall, Mixed, no memory (seed 42, gate†)

$i \backslash j$	1	2	3	4	5	6
1	87.6					
2	87.6	90.8				
3	0.0	0.0	82.0			
4	1.6	0.0	9.9	95.8		
5	0.2	0.0	7.7	95.8	95.5	
6	87.3	85.6	0.3	8.1	0.0	86.6

SBR-Recall, Mixed, 50-document memory (seed 42, gate†)

$i \backslash j$	1	2	3	4	5	6
1	87.6					
2	89.0	89.8				
3	78.3	46.2	81.2			
4	80.3	46.6	79.1	96.7		
5	79.1	47.9	79.2	97.0	97.7	
6	85.8	75.0	79.0	96.9	97.3	81.6

D_g is two to three orders of magnitude above the largest encoder bucket on every condition ($111\times$ to $1608\times$). The empirical Fisher of a token classifier is largest at the head whatever the task (0.22 against 0.005 for the next group on LayoutLMv3), so the weighting itself favours the head. The informative part is the encoder-only share: with the head excluded, weighted displacement sits in the early and middle layers on every document encoder and is smallest in the late layers. Representational drift measured by CKA, in contrast, grows with depth because it accumulates. The two instruments agree on the classifier and disagree on the late encoder layers, and the main text claims only what both support.

F.2. What the significance tests do and do not show

Takeaway. Neither test is evidence for the localisation or for architecture-generality; those rest on the descriptive agreement of the profiles and on the remedy ordering.

Two tests accompanied an earlier version of this analysis and we report them with their limits. (i) A Mann–Whitney test of the classifier against each other parameter group (six samples per group, Bonferroni $\alpha' = 0.0042$) rejects equality

Table 9. **SBR-Recall delta capacity on CIL-CORD** (fp32 gate runs, 10-epoch cap, session-restricted scoring): diagonal $R[j,j]$ (each session when learned), final row $R[5,j]$ and AA. Generated (`supp_delta.tex`).

		Diagonal $R[j,j]$					Final row $R[5,j]$					
Delta	Seed	1	2	3	4	5	1	2	3	4	5	AA
rank 8, Q/K/V, memory	42	95.3	92.1	61.5	68.0	88.2	95.4	90.5	63.2	65.1	88.2	80.5
rank 8, Q/K/V, no memory	42	95.3	91.6	64.9	67.7	90.3	95.4	91.9	66.4	68.4	90.3	82.5
rank 64, Q/K/V + dense, no memory	42	95.3	92.2	91.6	88.8	93.3	95.4	93.2	92.2	89.3	93.3	92.7
rank 64, Q/K/V + dense, no memory	7	92.0	92.8	93.5	93.1	93.7	92.4	92.8	94.8	93.1	93.7	93.4
rank 64, Q/K/V + dense, no memory	123	95.4	89.8	96.2	89.4	91.8	95.5	90.4	95.8	88.6	91.8	92.4

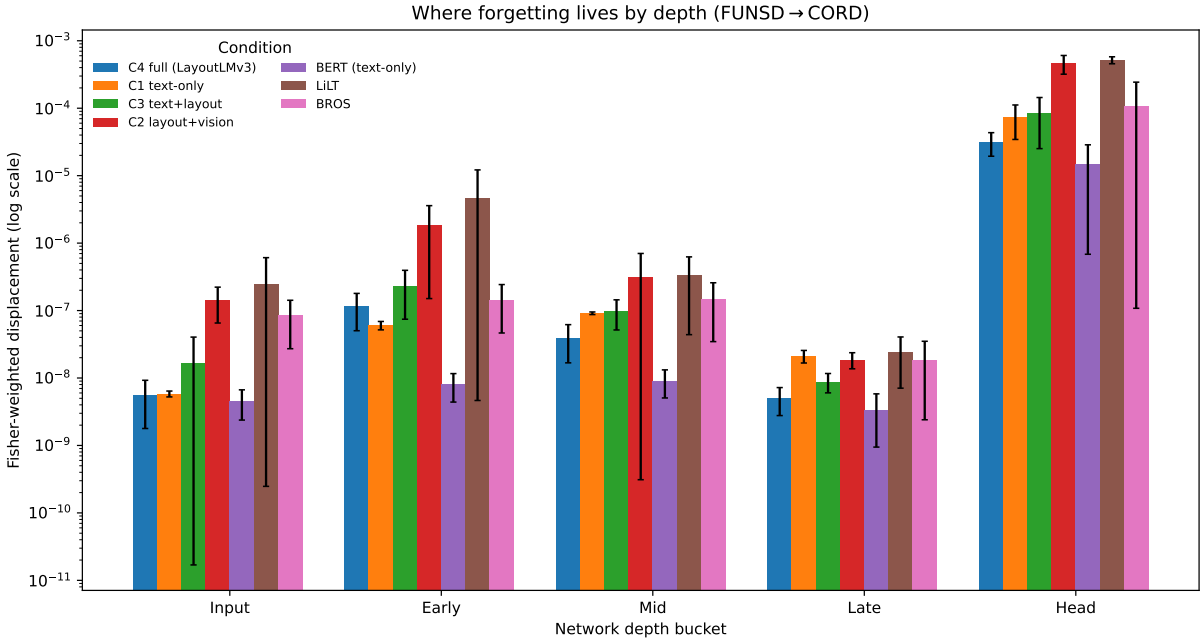


Figure 1. Where forgetting lives. Old-task-Fisher-weighted displacement D_g by depth bucket (log scale, first boundary), for the four LayoutLMv3 modality-mask conditions, text-only BERT, LiLT and BROS; mean over three seeds (two for text-only LayoutLMv3, whose third run is dead and excluded), ± 1 s.d. The classifier head's weighted displacement is two or more orders of magnitude above any encoder layer's on every backbone.

against three of twelve groups ($p = 0.0022$) and not against the rest; it was run on per-group Fisher *importance*, which is head-dominant by construction, and we do not use it as evidence of where forgetting occurs. (ii) A permutation test on the normalised D_g depth profile does not reject a shared profile between LayoutLMv3 and LiLT or BROS ($p = 0.38$, 0.51); since every normalised profile is nearly one-hot at the head, this test cannot reject and is not evidence of architecture-generality. The claim that drift is shared across backbones rests on the descriptive agreement of the CKA profiles and of Tab. 11, and on the remedy ordering of Tab. 18.

F.3. Consolidation probe

Takeaway. A uniform depth penalty prevents learning; the probe does not show that a targeted penalty recovers old tasks without replay.

A depth-targeted consolidation method (output distillation with a per-bucket penalty) scores 42.6 AA on DIL when the penalty is spread uniformly over all depth buckets, 84.7 when concentrated on the head and late layers, 86.1 head-only and 86.2 late-only (three seeds). All four arms include a 200-document head-replay buffer, and plain ER with the same buffer scores 87.9. We do not cite the probe for recovery without replay.

Table 10. **Controls that separate the label conflict from the memory** (CIL-CORD, LayoutLMv3, CoLaR memory with k=4, r=128, 50 documents per session unless stated; session-restricted scoring; AA per seed). Sliced: the cross-entropy is restricted to {O} and the classes that were live when the document was labelled (replay) or that are new (current "O" tokens). Rescore runs are fp16, gate runs fp32 (Tab. 17). Generated (`supp_conflict.tex`).

Row	Source	42	7	123	h/run
CoLaR, shared softmax	rescore†	28.8	31.9	–	1.4
CoLaR, shared softmax (gate run)	gate†	34.1	–	–	2.0
Replay loss sliced to classes alive at storage	rescore†	44.2	39.8	–	1.4
Replay loss sliced (gate run)	gate†	39.4	–	–	1.9
Replay and current losses sliced	gate†	62.9	69.2	49.4	2.3
+ model pseudo-labels written into the memory	gate†	48.0	–	–	2.1
+ pseudo-labels in both directions	gate†	44.1	–	–	2.4
True-label memory refresh (not deployable)	gate†	61.2	–	–	2.1
SBR, no memory	gate†	20.4	–	–	1.5
SBR, no memory, layers 1..11 frozen	gate†	14.8	–	–	1.0
CoLaR-SBR	gate†	87.6	86.3	90.1	2.4
Joint training (reference)	rescore†	92.8	90.5	92.4	0.4

Table 11. Fisher-weighted displacement at the first task boundary under naive fine-tuning (mean over the n live seeds; a run whose final AA is below 1.0 is dead and excluded, which removes one text-only LayoutLMv3 seed). "Head ratio": classifier D_g over the largest encoder bucket. "Encoder share": each bucket's fraction of the encoder total with the head excluded.

			Encoder share			
Condition	n	Head ratio	input	early	mid	late
LayoutLMv3, full	3	273×	0.03	0.70	0.24	0.03
LayoutLMv3, text+layout	3	361×	0.05	0.65	0.27	0.02
LayoutLMv3, text only	2	801×	0.03	0.34	0.51	0.12
LayoutLMv3, no text	3	247×	0.06	0.80	0.13	0.01
LiLT	3	111×	0.05	0.88	0.06	0.00
BROS	3	738×	0.21	0.37	0.37	0.05
BERT	3	1608×	0.18	0.32	0.36	0.14

F.4. Remedy benefit across backbones

Takeaway. The cause is shared by four encoders; the benefit of a remedy without old-task supervision is not, and can change sign.

Fig. 2 plots, per backbone and scenario, the change in AA over naive training of each classical remedy, from the earlier grid (Tab. 18; legacy scoring on CIL-CORD and Mixed). Replay's benefit is large and consistent; EWC and LwF change sign across backbones, which is why the main paper's drift table reports four backbones.

F.5. The falsification ledger

Takeaway. Five families that keep updating a shared network without stored old-task targets were each built in their strongest form we could manage, and each fails for a reason identified per boundary, not inferred from the score.

Tab. 12 lists them with their best AA on DIL and the failure reason; the last row is the label-refresh family of Appendix E.

F.6. What the memory must hold

Takeaway. Whole documents retain and every summary of the same store collapses; each document can then be compressed on its own (Figs. 3 and 4).

Routing which stored document to replay (null). The OCR vocabulary of a document is set by the input and does not drift with training, which suggests a drift-free sampler: a softmax over term-frequency signature scores $s_j = \sigma(x)^\top \sigma_j$ in three directions (*near*, *far*, *task*), uniform as the zero-point. The seed-42 reading was promising (far $+1.8$ over uniform 86.4 at d=50), so we fixed that single arm in advance for confirmation. Across three seeds the gap collapses to $+1.8/+0.4/-0.6$, mean $+0.5$ points with a sign flip, inside the uniform-sampling spread: lexical routing is a null, and the CoLaR results do not depend on it.

Rebalancing and reading the memory (nulls). On DIL with LayoutLMv3 (CoLaR, k=4, r=128, 50 documents per task; base 87.6 on seed 42), three changes to how the memory is filled or weighted leave AA where it was: a class-balanced replay loss gives 87.1, coverage-based (k-centre) document selection 86.8 and entity-weighted compression 87.1 (seed 42 each). Each trades FUNSD retention for SROIE retention at about 1:1, so the changes redistribute retention between tasks rather than add to it. A three-seed check finds no lever above the base (class-balanced 87.8, k-centre 87.0, against 87.8±0.5 for the base). Two read-side additions are negative: blending a kNN readout over the stored latents into the head is inert or harmful, and metaplastic consolidation of the readout lowers AA from 85.9 to 79.5 at its own operating point.

Budget-adaptive allocation is a frontier, not a gain. Allocating a fixed compressed-replay budget unevenly across tasks (documents $[18, 5, 2]$, ranks $[64, 128, 64]$ vs. a uniform 10 documents at rank 64; 5 epochs, k=4) gave $+4.0$ AA on seed 42 while passing a per-domain guard fixed in advance. On seeds 7 and 123 the same comparison gave -8.0 and $+5.7$: $+0.6 \pm 7.5$ over three seeds, a null. The seed-42 result was a point on a Pareto frontier (old-domain retention traded against the final domain at $\approx$ 9:1), not a shift of the frontier, and we report it as such.

A bounded positive on the storage side. Replaying pseudo-labelled *public* documents through the same mechanism (zero private bytes) recovers dense-label tasks (FUNSD 76.1) but not one the proxy pool does not cover (SROIE $\sim$ 7), topping out at 58.9: coverage-limited transfer.

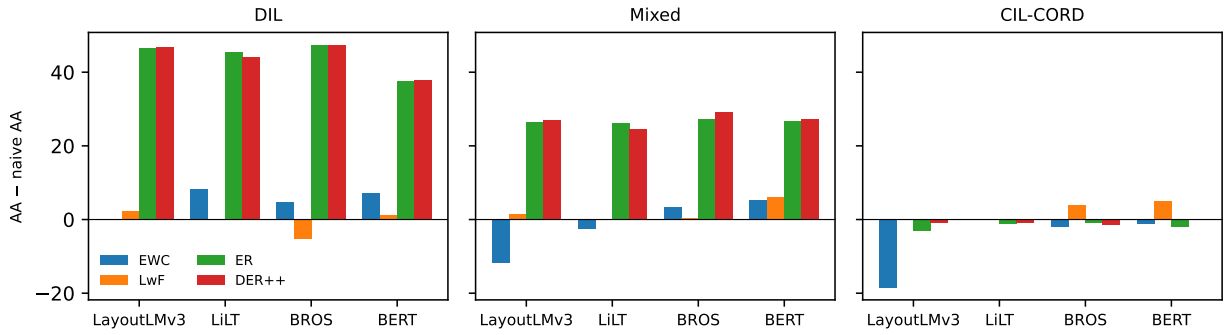


Figure 2. Remedy benefit is backbone-dependent; the phenomenon is not. Change in AA over naive for each classical remedy, per backbone and scenario (mean of three seeds, earlier grid, legacy scoring on CIL-CORD and Mixed). Replay's benefit is large and consistent; EWC and LwF change sign across backbones (missing bars: cells not run).

Table 12. Falsification ledger, DIL/LayoutLMv3 (naive 41.3, replay ≈ 88). Instance names are ours and belong to the family in the first column; LexSlot selects head slots by OCR-vocabulary overlap, and CoLaR-R is CoLaR with the stored labels refreshed. Each family is the strongest instance we could build of its mechanism; the failure reason is from a per-boundary root-cause analysis, not inferred from the score. †run at a frozen-backbone, short-schedule (3–5 epoch) operating point whose own naive level is given in the cell. ‡CIL-CORD and Mixed, LayoutLMv3, three seeds each, legacy scoring; pseudo = refresh with the model's own predictions, oracle = refresh with the true labels (not deployable).

Family	Instance	Best AA	Why it fails
Weight merging	TIES/Fisher head merge†	21.6 (no-merge 21.9)	merge of head vectors cancels; near no-op
Merge + realignment	LCA (ported)	43.9±2.5	backbone merge halves BWT; Gaussian realignment kills acquisition
Subspace transfer	HGT / CUBER (ported)†	39.8 (own naive 38.7); HGT ≤ 25.3	protected subspace grows; head unconstrained; α is a pure stability–plasticity dial
Parametric slots	LexSlot (no buffer)	42.2	isolated head slots cannot re-ground shared head
Input-anchored memory	Ledger†	FUNSD 21 / SROIE 23	own-vs-cumulative gap is training-free
Feature-Gaussian replay	LexMem v3b (positive control)	66.0	partial: recovers dense-label tasks only
	LexMem v5 (+relational)	63.2; mid-task 13.0	marginal summary breaks doc consistency
Label refresh (latent memory)	CoLaR-R pseudo (CIL-CORD / Mixed, 3 seeds each)‡	20.0 / 59.5 (plain 20.4 / 58.8; oracle 26.3 / 64.4)	teacher confident on ≈ 0 stale-O tokens; pseudo refresh is a no-op on both scenarios (oracle refresh +5.9 / +5.6 shows the slot is reachable only with true labels)

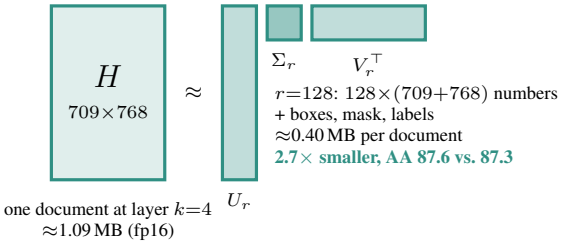


Figure 3. **CoLaR stores whole documents, compressed one at a time.** The pooled layer-k feature space is near full-rank (rank 16 keeps 43% of the variance), but a single document's activations are low-rank: its tokens share one layout and one vocabulary. Sizes for one padded document (512 text and 197 image tokens); AA on DIL, LayoutLMv3, seed 42, against uncompressed latent replay. Truncated SVD per document keeps every token's feature at its own position with its own label, which is the binding that every marginal summary destroys (Fig. 4).

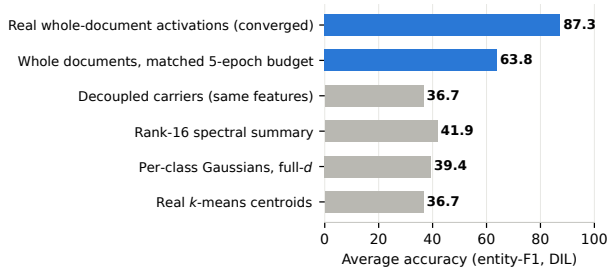


Figure 4. **What the memory must hold** (DIL, LayoutLMv3, layer k=8, AA, single-seed diagnostics). Whole documents retain; summaries of the same store fall to the naive level (41.3), and the identical features on decoupled layout carriers lose 27 points against matched whole documents.

F.7. Image class-incremental learning

Takeaway. On images the ranking inverts: a class-conditional Gaussian head beats the same latent memory, because an image carries one label and its readout is summarised by a per-class statistic; the trunk is intact and the gap is readout coverage.

Tab. 13 gives the fixed comparison set on ViT-B/16 (CIFAR-100, ImageNet-R; ten sessions) and Tab. 14 extends it with BWT and with every image run outside the fixed set that we keep as evidence, all under the same recipe (Appendix H). Two readings matter. Replay at 200 exemplars is far below the 2,000-exemplar rows (42.3/21.6 vs. 76.2/54.8 for ER), so the memory size in Tab. 13 is the one at which replay is competitive, not a favourable choice for our rows. SLCA's whole margin comes from its classifier alignment:

Table 13. **Fixed comparison set on image CIL** (ViT-B/16, ten sessions, AA, mean ± s.d., seeds 42/7/123): the same references (naive, joint training) and baselines as the comparison table of the main paper, with SLCA and RanPAC in the slots CPFD and IS3 take on documents. ER/DER++/SLCA keep 2,000 exemplars; the uncompressed latent memory and CoLaR keep 500/class. Ridge is a diagnostic readout refit, not a method.

Method	CIFAR-100	ImageNet-R
Naive	11.7 ± 1.1	9.3 ± 0.3
LwF	15.7 ± 0.6	12.7 ± 1.4
ER (2,000 exemplars)	76.2 ± 1.2	54.8 ± 0.7
DER++ (2,000 exemplars)	76.5 ± 1.5	52.8 ± 1.1
CODA-Prompt	15.2 ± 1.8	8.4 ± 0.7
CL-LoRA (2025)	14.1 ± 1.2	11.4 ± 0.8
SLCA (2023)	87.5 ± 0.2	65.0 ± 0.3
RanPAC (2023)	85.3 ± 0.1	58.5 ± 0.3
LCA (2026)	77.2 ± 3.2	54.7 ± 3.6
Latent memory, 500/class (ours, uncompressed)	80.0 ± 0.7	55.1 ± 1.3
CoLaR r=128, 500/class (ours)	79.5 ± 1.0	55.3 ± 1.4
Ridge readout on the latent-memory trunk (probe, seed 42)	88.3	72.3
Joint training (reference)	89.5 ± 0.4	77.6 ± 0.5

Table 14. Image class-incremental learning, AA and BWT, mean±std over seeds 42/7/123 (ViT-B/16, ten sessions). Fixed comparison set first, our rows, joint training (reference), then the dominated rows kept as evidence. SimpleCIL is a single seed.

	CIFAR-100		ImageNet-R	
Method	AA	BWT	AA	BWT
Naive	11.7 ± 1.1	-95.6 ± 1.4	9.3 ± 0.3	-87.1 ± 0.8
LwF	15.7 ± 0.6	-91.0 ± 0.7	12.7 ± 1.4	-84.4 ± 1.4
ER (2,000 exemplars)	76.2 ± 1.2	-23.8 ± 1.3	54.8 ± 0.7	-37.9 ± 0.9
DER++ (2,000 exemplars)	76.5 ± 1.5	-23.6 ± 1.7	52.8 ± 1.1	-40.5 ± 1.5
CODA-Prompt	15.2 ± 1.8	-89.7 ± 1.9	8.4 ± 0.7	-67.4 ± 1.7
CL-LoRA (2025)	14.1 ± 1.2	-92.7 ± 1.4	11.4 ± 0.8	-82.2 ± 0.8
SLCA (2023)	87.5 ± 0.2	-7.8 ± 0.2	65.0 ± 0.3	-13.5 ± 0.2
RanPAC (2023)	85.3 ± 0.1	-5.0 ± 0.1	58.5 ± 0.3	-7.0 ± 0.3
LCA (2026)	77.2 ± 3.2	-15.8 ± 3.8	54.7 ± 3.6	-21.8 ± 4.6
Latent memory, 500/class (ours, uncompressed)	80.0 ± 0.7	-19.8 ± 0.7	55.1 ± 1.3	-37.4 ± 1.3
CoLaR r=128, 500/class (ours)	79.5 ± 1.0	-20.4 ± 1.1	55.3 ± 1.4	-36.8 ± 1.5
Joint training (reference)	89.5 ± 0.4	–	77.6 ± 0.5	–
EWC	13.9 ± 1.5	-93.1 ± 1.6	9.3 ± 0.2	-87.2 ± 0.1
ER (200 exemplars)	42.3 ± 2.7	-61.6 ± 3.1	21.6 ± 1.3	-75.2 ± 1.4
DER++ (200 exemplars)	43.6 ± 4.0	-60.6 ± 4.2	20.2 ± 2.5	-77.2 ± 2.5
SLCA without classifier alignment	38.0 ± 3.9	-66.7 ± 4.3	25.8 ± 0.7	-67.4 ± 0.9
SimpleCIL (frozen trunk, NCM)	81.7 (1 seed)	-5.5 (1 seed)	48.2 (1 seed)	-6.2 (1 seed)
Per-class Gaussian latent replay (falsified)	10.7 ± 0.8	-96.5 ± 1.0	–	–

without it the same recipe falls to 38.0/25.8, below 200-exemplar replay on CIFAR-100, which is the image-side instance of the head-locus finding. Per-class Gaussian latent replay, the image port of the document-side Gaussian store, sits at the naive level (10.7).

Ridge decomposition of the plastic-head gap. Tab. 15 reports a ridge-readout probe on the trained trunk of our replay rows (seed 42). After the last session the trunk is frozen and a closed-form ridge readout (random projection to 10,000 ReLU features, λ=10, the RanPAC recipe) is fitted under three data conditions: on the stored memory only (deployable), on all training data of all sessions (a non-deployable ceiling on what the features support), and on all data through the frozen pre-trained trunk (image only). Three differences follow: the *head gap* (full ridge minus the trained head) is

Table 15. Ridge decomposition, seed 42, AA (LayoutLMv3 on documents, ViT-B/16 on images). *Head*: the trained plastic head. *Ridge, stored/full/frozen*: closed-form readout on the stored latents, on all seen training data, and on all data through the frozen pre-trained trunk. Gaps as defined in the text; a dash means the condition does not apply (documents have no frozen-trunk reference).

Cell	Head	Ridge, stored	Ridge, full	Ridge, frozen	Head gap	Trunk gap	Coverage gap
CIFAR-100, latent memory	78.8	80.8	88.3	85.4	9.5	2.9	7.5
ImageNet-R, latent memory	56.2	58.0	72.3	57.3	16.1	15.0	14.3
DIL, CoLaR	90.9	91.1	91.3	–	0.3	–	0.2
CIL-CORD, CoLaR	67.5	75.2	75.2	–	7.7	–	0.0
Mixed, CoLaR	70.9	71.0	71.6	–	0.7	–	0.6

the total headroom of the readout; the *coverage gap* (full minus stored) is the part that needs features of inputs the memory does not hold; the *trunk gap* (adapted minus frozen) is what replay-driven adaptation bought. On images the head gap is 9.5/16.1 points and is mostly coverage (7.5/14.3), so a better analytic head on the stored latents does not close the gap to SLCA; the adapted trunk is nevertheless 2.9/15.0 points better than the frozen one under the same readout, so adaptation is not the problem. On documents the head gap of CoLaR is within one point on DIL and Mixed and 7.7 points on CIL-CORD with zero coverage gap, i.e. the compressed store already covers its documents and the residual is the plastic head itself. The probe reads every past training image once at the end and is therefore a diagnostic, not a continual method.

G. Baselines and protocols

Baselines and ports. EWC uses λ=10^3 with the Fisher over 200 documents; LwF T=2, α=1; ER and DER++ a reservoir of 200 samples in the legacy cells, DER++ α=β=0.5. Our O-LoRA and CL-LoRA ports train a single rank-8 LoRA adapter on the query, key and value projections of the frozen pre-trained encoder, shared by all sessions and retrained in each with an orthogonality penalty (λ=0.5) against its own earlier states; CL-LoRA adds distillation towards the previous model (T=2). The published methods keep one adapter per task, so these rows report our ports, not the published designs. CPFD and IS3 are ported onto the single growing token-classification head used by every other method (the reference implementations keep per-task classifiers). LCA merges the backbone with TIES and realigns the classifier with Gaussian feature replay.

Prompt-, LoRA- and flatness-based baselines. Tab. 16 reports the frozen-backbone prompt family (L2P, DualPrompt, CODA-Prompt), the orthogonal/continual LoRA family and ER with C-Flat++ sharpness-aware optimisation on DIL, three seeds, LayoutLMv3 and BERT. The prompt family forgets little but never acquires the task (AA ≈ 30–35, below the naive level of 41.3): a frozen document encoder with a drifting shared head cannot be routed back into place.

Table 16. Prompt/LoRA/C-Flat++ baselines, DIL, AA and BWT, mean±std over seeds 42/7/123. Naive 41.3 (LayoutLMv3), ER 87.9.

	LayoutLMv3		BERT	
Method	AA	BWT	AA	BWT
L2P	30.4 ± 0.4	-11.5 ± 0.1	34.4 ± 0.7	-38.5 ± 1.0
DualPrompt	29.9 ± 0.6	-11.1 ± 0.2	34.0 ± 0.2	-40.2 ± 1.1
CODA-Prompt	32.5 ± 1.2	-10.5 ± 0.2	35.2 ± 0.7	-40.7 ± 1.3
O-LoRA	40.8 ± 0.3	-56.8 ± 1.0	37.4 ± 0.3	-53.4 ± 0.9
CL-LoRA	41.1 ± 0.3	-58.8 ± 2.1	38.6 ± 0.4	-31.6 ± 1.9
ER + C-Flat++	88.4 ± 0.1	-1.8 ± 0.6	78.3 ± 0.4	-2.2 ± 0.1

Our single-adapter LoRA ports acquire but forget as much as naive training does. Only the replay-based variant reaches the level of replay, and its sharpness term adds nothing over plain ER (88.4 vs. 87.9 on LayoutLMv3), consistent with Sec. 4.1 of the main paper: the remedy ordering is set by whether the head is re-grounded, not by the optimiser.

Settings of the legacy-protocol cells. Cells marked † come from three groups of earlier runs: the *gate* runs of our methods and controls (`results/gates/cil_protocol_2026-11`, fp32), the *rescore* runs of the baselines and of CoLaR on CIL-CORD and Mixed (`results/rescore`, mostly fp16), and the earlier *grid* (`results/<run>`). They differ from the clean protocol in three ways. First, every legacy cell early-stops on the session's *test* split, except joint training on CIL-CORD and Mixed, which holds out 10% of its pooled training set. Second, on CIL-CORD and Mixed the epoch cap differs by method: 100 for naive, joint training, ER, DER++, LwF, CPFD and IS3, 30 for CODA-Prompt and CL-LoRA, 10 for the CoLaR-family rows and LCA. Third, precision differs: most rescore cells ran in fp16 mixed precision, but IS3 and LCA always ran in fp32 (their training loops never enable autocast), and seed 42 of ER, DER++, LwF, CPFD, CL-LoRA and CODA-Prompt on CIL-CORD ran in fp32 because it started before mixed precision was wired into those loops, so several cells mix precisions across seeds. The CIL-CORD cells of CoLaR-SBR, no-memory SBR, SBR-Recall and LexSlot, and the DIL SBR cells, are fp32 gate runs. In the 2×2 the shared-softmax cells are therefore fp16 or mixed and the blocked-readout cells fp32. The first DIL grid saved no configuration; for it Tab. 17 gives the grid script's defaults, which could not be confirmed per run. Tab. 17 lists, for every group of † cells, the epoch cap, numeric precision, memory and seed count, read from each run's log and saved configuration. A cell never combines the two protocols.

Legacy-scored tables. Tabs. 18 and 19 are the tables of the earlier grid. Their CIL-CORD and Mixed columns use legacy scoring, under which joint training reads 33 on CIL-CORD; they are kept for the cross-backbone comparison (the EWC sign flip of Sec. 4.1 of the main paper is read from the Mixed rows of Tab. 18) and for completeness. "Independent" is the mean of one model trained per task from the pre-trained checkpoint.

Null cells among the fixed baselines (CPFD, IS3, CODA-Prompt). Three of the fixed baselines do not converge on the document scenarios, and we report those cells as nulls rather than as scores. The CIL-CORD and Mixed values for CPFD and IS3 below come from the earlier grid (fp32, 10-epoch cap, test-split early stopping, legacy scoring; Tab. 19); the session-restricted values in the main paper's comparison table come from later re-runs with a 100-epoch cap, which are different models, not a rescoring of these. On CIL-CORD, CPFD learns the first session (90–95 F1) and then reads 0.00 on every session from the first class-adding boundary (AA 2.7 and 0.0 on seeds 42 and 123); on Mixed it reads 6.0, 0.0 and 0.1; and on DIL, where no class is ever added, the first domain falls to $8.0/4.0/0.9$ F1 at the first boundary and the final row is zero on every domain for seed 42 and $3.0/9.7/0.0$ for seed 7 (AA $0.0/4.2/14.2$, reported as 6.1 ± 7.3 since one seed does learn the last domain), so the collapse is not a property of head growth. Two single-knob ablations on seed 42 reproduce the collapse exactly: flooring the per-sample CE weight at 0.5 (session-1 F1 0.00) and switching the distillation term off entirely (session-1 F1 2.1), so neither the adaptive CE factor nor the attention/logit distillation is the cause; what remains is the pseudo-labelling and O-token masking path, which removes the training signal on the new session's entities. IS3 learns each new session (diagonal $93/29/32/66/10$ on seed 123) but never predicts a previously learned class again (every off-diagonal entry is exactly 0.00), giving AA 4.4 ± 2.2 on CIL-CORD in the earlier grid (three seeds); on Mixed, whose last session adds no classes, it recovers there to 30.1 ± 1.2. The main comparison reports the session-restricted re-runs instead. On DIL the nine-tag head never grows, so IS3's old/new split is degenerate (every entity row is "old" at every boundary and the O-token distillation spans the whole output) and the method reduces to sequential fine-tuning under a full-output distillation term; seed 42 learns the first domain (88.2) and drops both earlier domains below 8 F1 at the next boundary (AA 38.3 vs. naive 41.3). We report that single seed rather than spend further compute on a structural limitation of the port. CODA-Prompt reads AA 0.0 on all three CIL-CORD seeds with its default pool of 10 components; enlarging the pool to 50 (10 per session) lets some sessions be learned (diagonal $79.5/40.6/0.0/18.9/36.8$) but the final row is $[0, 0, 0, 0, 36.8]$, AA 7.4, with one session never learned, so the cell stays a null. These are properties of the methods' O-tag and static-pool assumptions under five class-adding sessions, not of the sessions themselves: the

Table 17. **Settings of the legacy-protocol (†) cells.** Every row early-stops on the session's test split, except joint training on CIL-CORD and Mixed, which holds out 10% of its pooled training set. Precision "fp32/fp16": seed 42 on CIL-CORD ran in fp32 (mixed precision was added to these training loops only afterwards) and every other seed in fp16 mixed precision. Seeds are given per scenario in the order of the second column. The clean protocol replaces all of this with fp32, a 10-epoch cap (100 for joint training) and early stopping on 10% held out from each session's training data. ‡LwF: the one fp16 seed per scenario that collapsed to zero is excluded (CIL-CORD seed 123, Mixed seed 7), leaving 2 and 1 seeds. *The first DIL grid (naive, EWC, LwF, ER, DER++, CODA-Prompt, CL-LoRA, joint training; all four backbones) saved no configuration; the values are the grid script's defaults and could not be confirmed per run (that script's default batch size was 16, against 2 for every later run), and joint training early-stopped on the pooled test sets.

Row	Scenarios	Epoch cap	Precision	Memory	Seeds
Naive, joint training	CIL-CORD, Mixed	100	fp16	none	3, 2
ER, DER++	CIL-CORD, Mixed	100	fp32/fp16	200 samples (reservoir)	3, 2
LwF	CIL-CORD, Mixed	100	fp32/fp16	none	3, 2‡
CPFD	CIL-CORD, Mixed	100	fp32/fp16	none	2, 2
IS3	CIL-CORD, Mixed	100	fp32	none	2, 2
CODA-Prompt, CL-LoRA	CIL-CORD, Mixed	30	fp32/fp16	none	2, 2
LCA	CIL-CORD, Mixed	10	fp32	class Gaussians	2, 2
CoLaR	CIL-CORD, Mixed	10	fp16	50 docs/session	2, 2
CoLaR-SBR	CIL-CORD	10	fp32	50 docs/session	3
CoLaR-SBR	Mixed	10	fp16	50 docs/session	2
SBR without memory	CIL-CORD	10	fp32	none	1
CoLaR-R, sliced replay loss	CIL-CORD	10	fp16	50 docs/session	2
CoLaR, true-label memory refresh	CIL-CORD	10	fp32	50 docs/session	1
SBR-Recall	CIL-CORD, Mixed	10	fp32	none, 50 docs/session	3, 1
CoLaR-SBR + LexSlot	CIL-CORD, Mixed, DIL	10	fp32	50 docs/session	3, 3, 3
CoLaR, CoLaR-SBR, CPFD, IS3, LCA	DIL	10	fp32	CoLaR rows: 50 docs/task	3 (IS3: 1)
First-grid rows*	DIL	100 (CODA-Prompt 30)	unknown	ER, DER++: 200 samples	3

Table 18. Generality grid (legacy scoring on CIL-CORD and Mixed, test-split early stopping): average accuracy (mean ± s.d., three seeds) of the six classical strategies on four backbones and three scenarios. *Indep.* is a per-dataset reference (one model trained independently on each dataset, three-seed macro over FUNSD/SROIE/CORD). *Joint* is joint training on the pooled data (one model, all three datasets), also a reference. Independent is only meaningful on DIL, where the task stream is three distinct datasets; CIL-CORD is one dataset split by label and Mixed is a composed stream, so Indep. is undefined there and marked N/A. Grey cells did not fit the 6 GB GPU. On DIL replay comes close to joint training on every backbone; regularisation and distillation stay within ten points of naive training on average; on CIL-CORD every method stays below joint training under this scoring, and only LwF (on BROS and BERT) rises above naive training.

Scenario	Backbone	Naive	Indep.	Joint	EWC	LwF	ER	DER++
DIL	LayoutLMv3	41.3±1.5	88.8±0.6	88.7±1.1	41.2±1.1	43.7±1.1	87.9±1.4	88.0±0.9
	LiLT	40.1±0.5	85.3±0.8	83.8±1.2	48.4±5.9	–	85.7±0.5	84.4±0.3
	BROS	40.3±1.8	88.8±0.1	87.5±1.3	45.0±4.9	35.1±3.1	87.8±0.4	87.5±1.1
	BERT	39.4±0.3	77.3±0.5	77.2±0.4	46.6±3.0	40.5±0.1	77.1±0.9	77.4±0.6
CIL-CORD	LayoutLMv3	19.0±0.2	N/A	33.3±0.3	0.5±0.6	18.8±0.2	16.0±1.4	18.2±0.3
	LiLT	19.1±0.3	N/A	32.4±0.4	–	–	17.9±0.5	18.0±1.0
	BROS	19.2±0.1	N/A	32.9±0.0	17.3±0.8	23.1±1.2	18.3±1.5	17.7±1.1
	BERT	19.3±0.1	N/A	32.4±0.2	17.9±0.3	24.3±0.3	17.1±1.6	19.2±1.9
Mixed	LayoutLMv3	33.9±0.3	N/A	67.5±0.8	22.0±9.4	35.2±1.2	60.3±5.3	60.8±2.7
	LiLT	30.7±0.7	N/A	61.8±0.6	28.1±3.0	–	57.0±1.4	55.1±2.9
	BROS	33.9±0.4	N/A	65.3±2.1	37.4±3.3	34.4±0.5	61.1±1.4	63.0±0.4
	BERT	23.4±0.3	N/A	55.0±0.6	28.6±0.9	29.4±4.2	50.2±0.6	50.7±1.3

same sessions are learned by naive training (AA 19.0) and by joint training (33.3), both under legacy scoring.

Per-backbone forgetting. The full BWT and AF tables for every backbone×scenario×method cell of the generality grid are generated from the run directories and will ship with

Table 19. **Fixed comparison set on LayoutLMv3 under legacy scoring** (AA, mean ± s.d. over seeds 42/7/123, legacy protocol): naive and joint training as references and the published baselines (CPFD and IS3 in the slots SLCA and RanPAC take on images), then our rows; the image-side table is Tab. 13. *CoLaR-R, sliced*: CoLaR with the replay loss restricted to the classes alive when each document was stored. Its CIL-CORD and Mixed columns are runs of the earlier grid, not the † cells of the main paper: CoLaR, CoLaR-R, LexSlot, CPFD, IS3 and LCA ran in fp32 with a 10-epoch cap; the other rows saved no configuration (grid default cap 100, 30 for CODA-Prompt). The DIL column is the main paper's. LexSlot is the 200-exemplar slot-memory recipe. *did not converge on these sessions, which naive (19.0) and joint training (33.3) do learn ("Null cells among the fixed baselines" below). "–": not applicable; IS3 on DIL is a single seed (fixed-label port limitation, see below).

Method	CIL-CORD	DIL	Mixed
Naive	18.98 ± 0.17	41.31 ± 1.54	33.85 ± 0.33
LwF	18.82 ± 0.15	43.66 ± 1.05	35.23 ± 1.17
ER	15.98 ± 1.36	87.85 ± 1.42	60.32 ± 5.25
DER++	18.15 ± 0.30	**88.02 ± 0.88**	**60.82 ± 2.74**
CODA-Prompt	null*	32.51 ± 1.23	8.11 ± 1.52
CL-LoRA (2025)	15.17 ± 3.14	41.14 ± 0.28	31.37 ± 0.63
CPFD (2023)	null*	6.15 ± 7.29	null*
IS3 (2024)	null*	38.29 (1 seed)	30.10 ± 1.18
LCA (2026)	8.15 ± 2.59	43.87 ± 2.45	19.01 ± 3.68
Text-only comparator (no layout or image):			
BERT-base	19.26 ± 0.11	39.42 ± 0.26	23.37 ± 0.27
CoLaR (ours)	20.38 ± 1.34	87.75 ± 0.48	58.75 ± 1.10
CoLaR-R, sliced (ours)	25.38 ± 2.38	–	61.30 ± 1.91
LexSlot (ours)	19.28 ± 0.12	87.33 ± 1.03	35.49 ± 1.39
Joint training (reference)	33.32 ± 0.33	88.73 ± 1.13	67.51 ± 0.82
Independent (one model per task)	–	88.79 ± 0.55	–

the code; Tab. 18 carries the AA view.

H. Reproducibility

Hyperparameters. Documents: AdamW, learning rate 5×10^{-5} (constant), weight decay 0.01, gradient clipping 1.0, batch size 2 with gradient checkpointing, early stopping with patience 2 and δ=0.1 F1 points with best-weight restoration. CoLaR family: k=4, r=128, 50 documents per session, 4 replayed documents per step, replay weight 1. SBR: threshold τ=0.5, all-background weight 1. SBR-Recall: rank 64, α=128 on query, key, value and dense layers above k (capacity ablation: rank 8, α=16, query/key/value). Baselines as in Appendix G. Image CIL (CIFAR-100, ImageNet-R; ViT-B/16, ten sessions): every reported cell, including the frozen-backbone and replay baselines, was trained on a 6 GB GPU at batch size 16 in fp32 with gradient checkpointing, 100-epoch cap with early stopping (patience 2), seeds 42/7/123; the CODA-Prompt, CL-LoRA and LCA image cells use this recipe unchanged.

Seeds and hardware. Seeds 42, 7 and 123 for every three-seed cell; a cell with fewer seeds shows its per-seed values. The clean runs, the gate runs and the rescore runs ran one at a time on a single NVIDIA RTX 2060 (6 GB) with about 14 GB of host memory; some rows of the first DIL grid ran on a rented GPU container (Tab. 17, first-grid rows).

Wall-clock. Tab. 6 gives the mean training-loop time per run of every headline cell, read from `metrics.json` (the sum of the per-session training times). The queue logs (`results/clean_2026-11/queue.log`, `results/gates/cil_protocol_2026-11/queue.log`) stamp the start and end of each job, which adds model loading and the final evaluations; for the clean CoLaR-SBR run of seed 42 the job took 2 h 01 min against 1.95 h of training loop. On CIL-CORD a CoLaR-SBR run takes about 2 to 2.5 hours of training loop on this card and an SBR-Recall run without memory about 1.5; the joint-training reference takes well under an hour because it is one pooled run.

What will be released. The code (methods, scenarios, the training entry point and the queue scripts), the configurations, and the per-run `metrics.json` and `matrix.npy` files from which every table and figure of the paper and of this appendix is generated, together with the two generators.